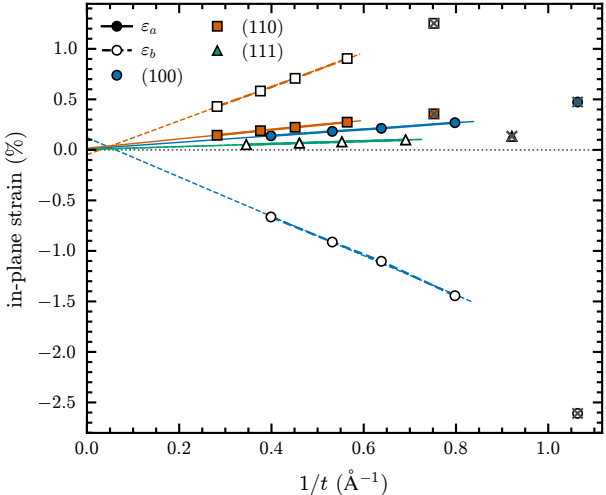




In-plane lattice distortion of freely relaxed (vc-relax) slabs against inverse bulk-equivalent thickness, solid for the a axis and dashed for b. Each curve is linear in $1/t$ and extrapolates to within 0.12% strain at infinite thickness, with fit residuals below 0.012%: the distortion is a surface effect that vanishes in the bulk limit, as it must, which is what validates treating these slabs as models of a surface rather than of a thin film. The 6L rung (struck through) is excluded from the fits, as it is from the thickness ladder: it lies off the $1/t$ trend, and keeping it would push the largest intercept to 0.64% and the largest residual to 0.150%. The facets differ in sign structure, and that is the physical content: on (100) the two in-plane axes move in opposite directions -- a expands while b contracts -- which is a shear, the lattice-space signature of the transverse-E channel. (111) is exactly isotropic, its two axes equal to numerical precision, so it can only shift D and never split E. (110) expands on both axes but by unequal amounts, so it carries an anisotropy without a sign change. These relaxations also fix the sign of tau independently of any convention: every axis carrying positive tau infinity (F1(b), or tau infinity.csv directly) expands here and the one carrying negative tau contracts, so positive tau is a compressive surface stress pushing the lattice outward. Strains are relative to each slab's own starting cell at the production a0; positive is expansion. Lines are least-squares fits in $1/t$ and the only extrapolation in this figure.